

Test 3: Earth's Dynamic Surface

Units 12–14 · pages 164–211 · 50 points · 50 minutes · Closed book

Name _____

Date _____

Score / 50

Read each question carefully. Circle one answer for multiple choice. Write on the lines provided; use the back of the page if you need more room. Point values are shown for each section.

Part A · Multiple Choice

12 questions × 2 points = 24 points

1. Exfoliation results from
 - a) the freezing of water inside cracks
 - b) the growth of plant roots
 - c) an increase in confining pressure as rock is buried
 - d) a decrease in confining pressure as overlying rock is removed, allowing the rock to expand and crack
2. Which mineral is virtually unaffected by chemical weathering?
 - a) Quartz
 - b) Olivine
 - c) Feldspar
 - d) Calcite
3. Caliche, a concrete-like soil layer found in hot arid regions such as the Mojave Desert, forms from
 - a) the leaching of clay and iron out of the upper horizons
 - b) frost wedging of the underlying bedrock
 - c) the accumulation of decayed organic matter
 - d) the movement of calcium ions and precipitation of calcite within the soil
4. Mass wasting occurs when
 - a) the gravitational force acting on a slope exceeds its resisting force
 - b) a slope is exactly at its angle of repose
 - c) chemical weathering outpaces mechanical weathering
 - d) the water table drops below the bedrock
5. P-waves slow at the mantle–core boundary and S-waves stop there entirely. Together these observations show that the outer core is
 - a) liquid
 - b) made of ordinary rock
 - c) cooler than the mantle
 - d) solid and highly rigid

6. The asthenosphere is best described as
 - a) the outermost layer of the crust
 - b) brittle and rigid
 - c) partially molten material that behaves plastically and can flow
 - d) entirely liquid metal
7. The puzzle of “apparent polar wander” was ultimately explained by concluding that
 - a) the magnetite crystals in the rocks were unreliable
 - b) Earth’s magnetic field switched off periodically
 - c) there were once two separate north magnetic poles
 - d) the magnetic north pole stayed fixed while the continents moved
8. The Ring of Fire around the Pacific Ocean is dominated by
 - a) divergent boundaries
 - b) hotspots
 - c) subduction zones
 - d) continent-to-continent collisions
9. Which of the following is an example of *intraplate* activity?
 - a) The San Andreas Fault
 - b) The Mid-Atlantic Ridge
 - c) The Yellowstone hotspot
 - d) The Himalayan mountains
10. Tension is the major type of stress found at which kind of plate boundary?
 - a) Convergent
 - b) Divergent
 - c) Transform
 - d) Intraplate
11. A fracture in rock along which there has been *no* movement is called
 - a) a fault
 - b) a joint
 - c) a fold
 - d) a slip
12. When tensional stress breaks crust into blocks that slide up and drop down along normal faults, producing alternating mountains and valleys, the result is called
 - a) a continental arc
 - b) a basin-and-range
 - c) an anticline
 - d) an island arc

Part B · Matching

1 question × 8 points = 8 points

13. Write the letter of the correct definition next to each term.

- | | |
|---------------------------------|--|
| _____ 1. Exfoliation | a) The supercontinent Wegener proposed once joined all the continents |
| _____ 2. Angle of repose | b) The steepest angle a cohesionless slope can hold without failing |
| _____ 3. Asthenosphere | c) The initial point within the crust where rocks rupture in an earthquake |
| _____ 4. Pangaea | d) Cracking of rock as confining pressure decreases and the rock expands |
| _____ 5. Subduction | e) Water-saturated sediment behaving like quicksand during shaking |
| _____ 6. Elastic rebound theory | f) The partially molten mantle layer that flows plastically |
| _____ 7. Focus | g) The sudden release of energy stored in deformed rock |
| _____ 8. Liquefaction | h) A denser plate plunging beneath another and down into the mantle |

Part C · Short Answer

4 questions × 3 points = 12 points

14. Explain why rocks that formed deep within the crust are so vulnerable to weathering once they reach the surface.

15. Name the three types of plate boundaries. For each, give the dominant type of stress and one real-world example.

16. Describe how convection in the mantle drives the motion of tectonic plates.

17. Explain why shallow earthquakes cause more damage than deep ones, and why scientists and the media report the epicenter rather than the focus.

Part D · Extended Response

1 question × 6 points = 6 points

18. A geologist studying a mountain range records four observations: the sedimentary layers are intensely folded, there are numerous thrust faults, large earthquakes are frequent, and there are no active volcanoes anywhere in the range. Identify the type of plate boundary that produced this range, explain each of the four observations, and name a real mountain range that fits.

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